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An Enhancement Process for Multi-focus Images Resulted from Image Fusion using qshiftN DTCWT and MPCA in Multiresolution Domain

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Abstract

Images having different focus values are combined using image fusion methodologies. The fused image shows superiority in the quality and highly informative than compared o any other multifocus images. The fused image would show its suitability for visual perception object detection due to its high quality than compared to multifocus images. Among the various quality parameters of the fused image, directional invariance and shift invariance significantly influence the fused image quality. The traditional image fusion methods using wavelets severely suffers with poor invariance and lack of directionality. An efficient image fusion method is developed using DTCWT with qshiftN and MPCA algorithms in the multiresolution (MR) domain. Multifocus input images are decomposed into high and low frequency components using MR algorithm. The decomposed frequency components of the input images are fused using DTCWT with qshiftN algorithm. The proposed fusion algorithm preserves both sift invariance and directional properties of the multifocus source image. Lastly, the fused image is processed using MPCA algorithm to enhance the features. The proposed methodology is assessed using various multifocus images and the evaluated metrics are contrasted with technologically advanced algorithms reported recently. The statistical metrics evaluated for the proposed method for different multifocus images shows superior properties than compared to the recently reported multifocus image fusion algorithms.

Keywords: Image Quality; qshiftN-DTCWT; Multiresolution; MPCA; Multifocus Image Fusion; Quality Evaluation Metrics

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1. Introduction

The crucial stage involved in various applications of image processing is an acquisition of images. The limitation involved in capturing an image is to focus all the objects. The optical lenses used in the acquisition of images have limited depth of field (DOF). The distance from the objects appeared in the image to optical lens usually creates multifocus images. To improve the DOF and to focus all the objects of the image and to improve their sharpness, an inexpensive methodology/algorithm needs to be developed. The developed technique should appreciably improve the focus of all objects by integrating the multiple images captured using different focal planes. Image fusion is a low-cost methodology used by various researchers in the last decade. The main objective of the different algorithms used in the image fusion process is to combine two or more images acquired in different focal planes. The obtained fused image should invariably show superior performance in the detection of objects than the objects that appeared in the multifocus images [1-3]. Since the social optical arrangement handles multiresolution knowledge under the transform field method computing principle, transform field algorithms, or more precisely, multiresolution algorithms, are superior. Numerous multiresolution approaches, including pyramid approaches, have been advanced in the literature [4-5], including distinct wavelet transformation [6], stationary transform wavelet [7-9], multiresolution singular value decomposition [10], DCHWT [11], lifting schemes of WT [12-13], and DDDWT [14] and ST [15]. Pyramid based approaches are generally suffered with the lack of selectivity in spatial familiarization. On the other hand, DWT based fusion algorithms helps to avoid these drawbacks. However, the DWT algorithm shows the lack of changing invariance and directional selectivity.

The object quality fused is defined by its shift invariance and directionality properties. The ringed objects within merged images that characterize the use of DWT for image fusion are present in traditional wavelet-based fusion algorithms. The DTCWT [16] is an example of an accurate wavelet transformation that can easily correct DWT's invariance and directionality deficiencies. However, since it is essential to follow both biorthogonal and stage requirements in DTCWT, the filter design method should be utilized. The qshiftN DTCWT [17] is a method employed in the design of filters that results in more consistent fusion efficiency. The only way to improve fusion is to use DTCWT. As an effective multiresolution transform for image fusion, qshiftN DTCWT has demonstrated its caliber to monitor complex directional and uniform characteristics.

The objective of the proposed approach is to create a high-quality merged image. High-quality images have more detailed object recognition, enhanced sharpness, better visual perception, and are free from distortions and noise. Users can easily perceive details in these fused images. A vast number of image fusion algorithms show its inadequateness in maintaining spatial resolution of the images, which generally causes blurring. In the proposed image fusion algorithm, the qshiftN DTCWT approach shows significant impact on enhancing the performance of the algorithm. This technique effectively enhances the resolution of fused images and yields high-quality results. The MR [18] and MPCA [19-20] approaches also suppress the influence of additive noise, distortion, and image edge preservation and other crucial parameters such as sectors with higher contrast images. From the visual and quantitative metrics, it is evidenced that the proposed methodology removes the distortions and artifacts. The proposed method is contrasted using qualitative and quantitative metrics with the methods described in the literature [21-22].

2. The Proposed Fusion Approach

The different stages involved in the proposed image fusion process are mentioned below:

S. No	Stage	Stage details	Purpose
1.	Stage 1	**Noise Elimination Process**	Convert a two-dimensional array to a one-dimensional array. By inverse the technique, the 2-D image could be restored from the 1-D vector data
2.	Stage 2	**Multiresolution DCT**	DCT is used to process the vector data. Consider the 1st half of the DCT coefficients to be LF and the rest to be HF. The LF coefficients are processed via IDCT to acquire vector data for the succeeding stage of disintegration [18].

3.	Stage 3	**qshiftN DTCWT**	A one dimensional, highly sampled DWT suffers greatly with invariance and shows poor directionality. These issues significantly overcome with the selection of DTCWT algorithm. The DTCWT algorithm shows shift invariant, computing efficient and directional selective. This algorithm generates actual and imagined components of the transformed coefficients with the help of wavelet filters [23-24]. The DTCWT is associated with two channel FIR filter banks. The actual and imagined components of the image is obtained from Tree A and Tree B of the filter banks respectively.
4.	Stage 4	**Modified Principal Component Analysis (MPCA)**	MPCA is used to turn uncorrelated variables into correlated variables. This method is useful for analyzing data and determining the optimal features for data collection.

2.1. qshiftN DTCWT

The DTCWT employs two significantly sampled filter banks, with a 2^d redundancy for a d-dimensional object. Figure 1 depicts the structure of a 1-D DTCWT filter bank in three stages. DTCWT-merged images are soft and unbroken, whereas DWT-merged images exhibit broken borders. DTCWT also provides great directional selectivity because it produces 6 sub-bands in each (±15∘, ±45∘, ±75∘), both real and imaginary, whereas DWT only gives restricted directions in (0∘, 45∘, 90∘), which increases transformational correctness and maintains more in-depth features. However, there are several issues with DTCWT's odd/even filter solution [24]:

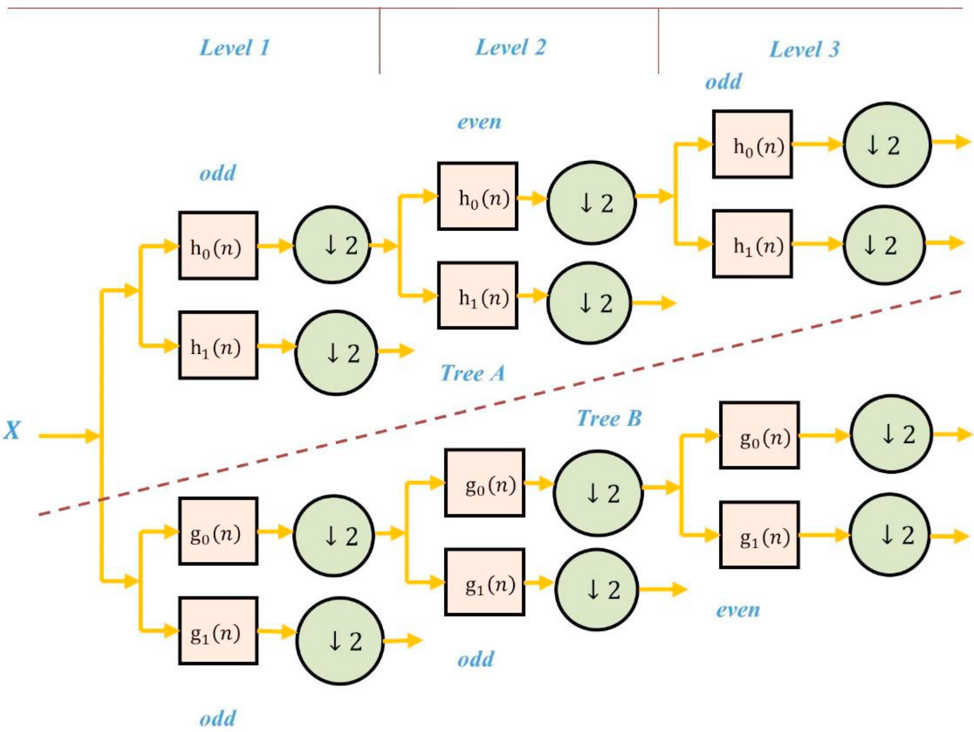

Fig.1. DTCWT bank filter's structure

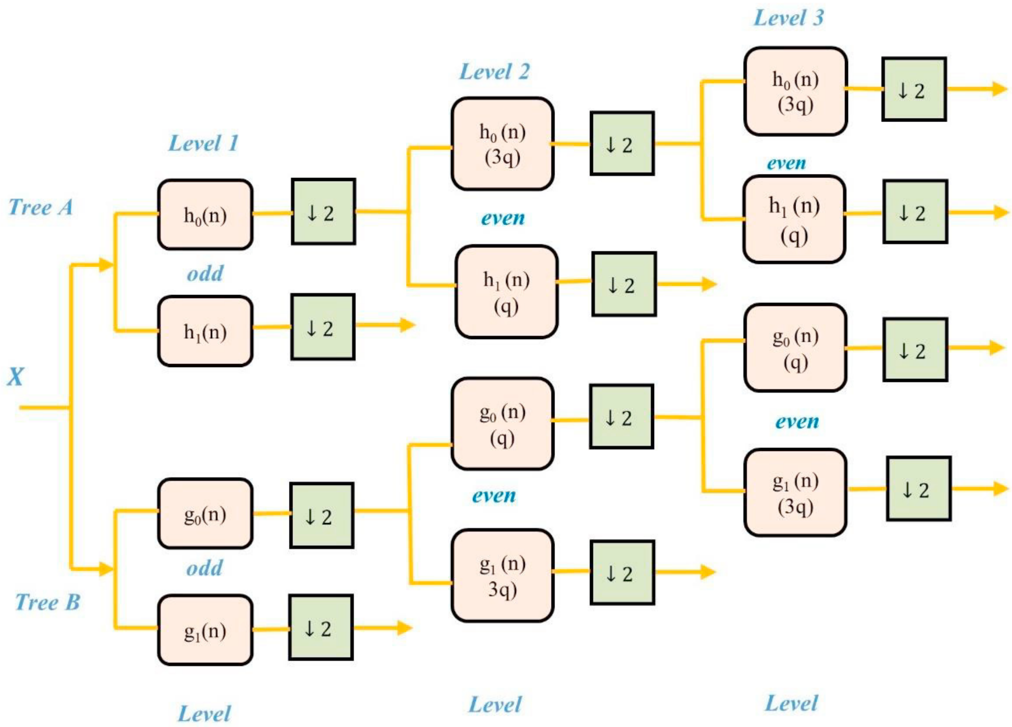

Fig. 2. The qshiftN DTCWT filter's structure

2.2. Modified Principal Component Analysis (MPCA)

The first principal component represents data with the greatest variance. Others are just as much of what's left. The data is well represented by the first principal component, which also illustrates the direction of maximum variation. In this paper, the MPCA approach is used to determine the best-represented value of each sub-band of source images after implementing the MR-based qshiftN DTCWT method. These values are then multiplied by matched source image sub-bands. MPCA's goal is to transfer data from the original space to the eigen space. By saving the components with the largest eigenvector, the variance of the data is enhanced and the covariance is lowered. Specifically, this method removes redundant data from source images and extracts the most significant components. Furthermore, MPCA prioritizes components with the greatest impact and resistance to noise. As a result, the MPCA decreases blurring and spatial distortions. The steps of the MPCA algorithm are as follows:

1. Create a vector from the data
2. Determine the covariance matrix of the given vector
$$\left(i.e., cov([im1(:)])\right)$$
3. Calculate the eigen values and eigenvectors of the covariance matrices
$$\left(ie., [V, D] = eig(C)\right)$$
4. Choose the first principal component in the order of the eigenvectors
$$\left(i.e., [max, ind] = sort(diag(D), 'descend')\right)$$
$$a = V(:, ind(1))./sum\left(V(:, ind(1))\right)$$
5. Finally, to get the features extracted image $\left(i.e., F\_E\_image = \ a(1) * im1\right)$

2.3. Flow Diagram of Proposed Approach

Figure 3 shows the detailed flow diagram of the complete fusion algorithm, which comprises two processes: MR-based qshiftN-DTCWT algorithm for image fusion and MPCA for feature enhancement. The fusion algorithm in the MR domain based on the qshiftN DTCWT image fusion mechanism and MPCA denotes the fusion of row components and column components of source images with a multiresolution, as shown in Algorithm 1.

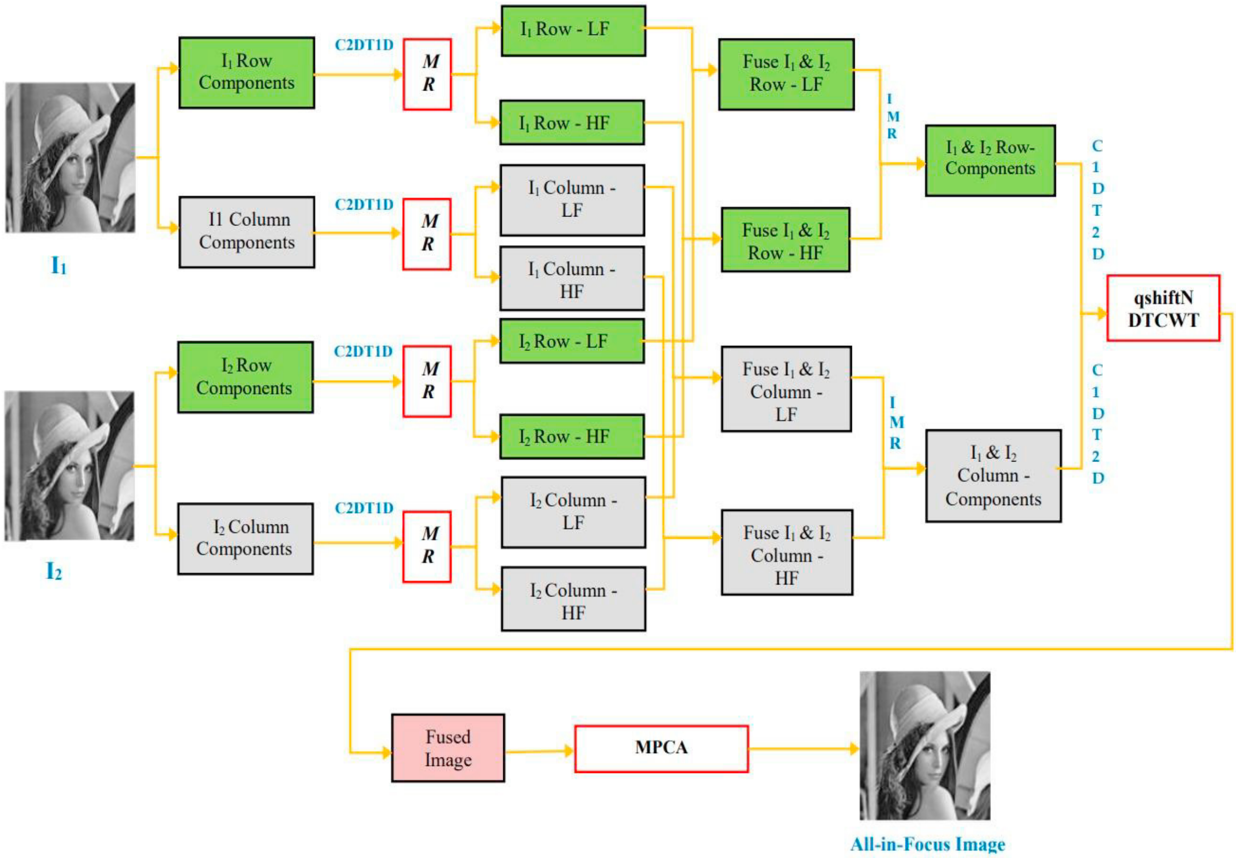

Fig. 3. The flow diagram of the proposed qshiftN DTCWT-MR and MPCA based image fusion algorithm

Algorithm 1: Various stages in the process flow of the proposed methodology is mentioned below:
Steps:

1. Take the multifocus images from the source and load them.
2. To use the image fusion technique, two multi-focus images (I1 and I2) are used as source images. Row (I1 and I2) and column (I1 and I2) pixels are used to divide raw images.
3. Multi-focus image row and column pixels are converted from a Two-Dimensional image to One-Dimensional array data.
4. Multiresolution is used to divide the resulting 1-D array data (I1) into minimum (row and column frequencies) and maximum (row and column frequencies) frequency elements. The I2 image is split into low (row and column frequencies) and high (row and column frequencies) frequency components in the same way.
5. The primary fusion procedure is performed to row elements (both low and high-frequency elements) of I1 and I2. Similarly, to generate minimum and maximum frequency column elements, this fusion technique is used to process the column elements.

6. To produce row and column elements, the merged row, and column frequency components are filtered utilizing the Inverse multiresolution algorithm.
7. The row and column elements of the 1-D array data are converted into a 2-D image.
8. Using qshiftN DTCWT, a final output image is created from the filtered row and column frequency elements.
9. The fused image is processed using MPCA algorithm to extract image features.
10. Final output image with extracted image features i.e., all-in-Focus image.

2.4. Evaluation of the proposed method's effectiveness

The performance of the proposed methodology is compared with high performance fusion techniques reported in the literature [21-22] using two categories: subjective assessment and objective evaluation. The subjective evaluation of the fused image is assessed based on the visual quality of the fused image. In many times the subjective evaluation is time consuming and expensive process. This led to raising a high demand to objective measures using various quality metrics to assess the performance of the proposed algorithm. This quantitative approach, known as objective analysis, is founded on mathematical modeling. It compares the spectral and spatial similarities of the fused image to the input images. There are two techniques for quantitative analysis: with and without a reference image. [25-48]. In this paper, seven evaluation metrics such as E(F), AG, CC, QAB/F, SSIM, QE, and QW is chosen to compare the proposed methodology with the work reported earlier.

3. Experimental Results

Quality measures such as E(F), AG, CC, $Q^{AB/F}$, SSIM, QE, and QW are employed to assess the algorithm's quality. These metrics are used to contrast the performance of the proposed methodology to the fusion methods reported earlier. The resemblance and robustness of the fused pictures against distortions are measured using these criteria. Standard source images are used in multi-focus image fusion for comparison. Experiments are also carried out on many images from various areas and datasets [49]. In these images, the proposed approach yields good results. However, these images are not included in the paper because the techniques that are contrasted with the suggested approach do not produce outcomes for these images. The statistical measure of the proposed methodology is evaluated with Desk, clock, lab, and flowerpot images and compared with the methodologies reported in the literature [21-22].

3.1. Comparison of multi-focus images

The evaluation of multifocus images (clock, desk, lab, flowerpot) using the proposed algorithm is compared with traditional methodology in the reported recently. Figures 4 to 7 represent the evaluation of multi-focus images (clock, desk, lab, flowerpot) using the proposed method. These figures reveal the original source images, left and right focus images and output fused images from the proposed methodology. The performance of the proposed method is evaluated using different metrics such as E(F), AG, CC, QAB/F, SSIM, QE, and QW. When it comes to quality measurements, the fused image has more information and clarity. Lastly the performance of the proposed methodology and the quality of the fused image is assessed using quality metrics. When compared to the approaches reported earlier [21-22], the proposed methodology is more successful in obtaining a high-quality fused image and the magnitude of the statistical parameters is shown in table 1 to 4 corresponding to various multifocus images and the best results of the methods are highlighted in bold.

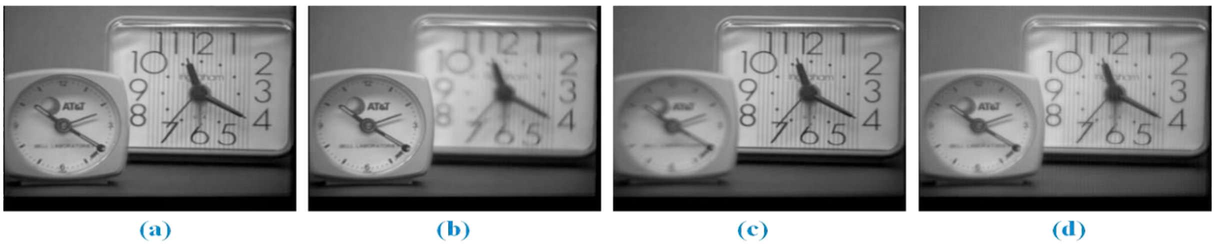

Fig. 4. (Clock): a. Original Image, b, c. Multifocus Input Images, d. Proposed Fusion

Table 1. The outcomes for the clock image and comparisons with existing techniques in the literature ([21]).

Evaluation Metric	***SSIM***	AG_F	$Q^{AB/F}$	***CC***	Q_E	***E(F)***
PA-DCPCNN	0.903	6.072	0.897	0.981	0.854	7.385
NSC -PCNN	0.9	5.411	0.873	0.981	0.803	7.304
BRW-TS	0.896	5.757	0.89	0.979	0.854	7.078
LG-MW	0.894	5.86	0.88	0.978	0.84	7.067
GFDF	0.895	5.802	0.892	0.978	0.853	7.077
MST-SR	0.903	5.879	0.893	0.981	0.85	7.321
CNN	0.896	5.751	0.891	0.979	0.855	7.128
DCT-CV	0.894	5.81	0.869	0.978	0.841	6.992
IM	0.9	5.746	0.85	0.978	0.848	7.17
BF	0.894	5.774	0.881	0.978	0.848	7.008
GF	0.897	5.841	0.895	0.98	0.852	7.249
CSR	0.897	5.382	0.889	0.979	0.85	7.318
QT	0.894	5.866	0.88	0.978	0.848	6.986
SSDI	0.896	5.848	0.893	0.979	0.853	7.123
DSIFT	0.894	5.857	0.894	0.978	0.85	7.014
Proposed Method	**0.959**	**6.23**	**0.994**	**0.986**	**0.899**	**7.428**

Fig. 5. (Desk): a. Original Image, b, c. Multifocus Input Images, d. Proposed Fusion

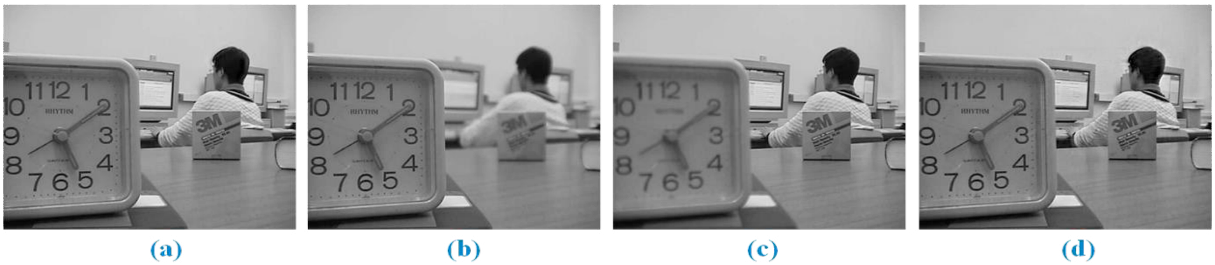

Fig. 6. (Lab): a. Original Image, b, c. Multifocus Input Images, d. Proposed Fusion

Fig. 7. (Flowerpot): a. Original Image, b, c. Multifocus Input Images, d. Proposed Fusion

Table 2. The outcomes for the desk image and comparisons with existing techniques in the literature ([21]).

Evaluation Metric	*SSIM*	*AG_F*	*$Q^{AB/F}$*	*E(F)*	*Q_E*	*CC*
BRW-TS	0.858	7.973	0.891	7.284	0.865	0.961
GFDF	0.856	7.995	0.891	7.279	0.864	0.961
SSDI	0.857	8.053	0.892	7.308	0.865	0.961
GF	0.862	7.921	0.893	7.309	0.866	0.962
BF	0.856	7.897	0.874	7.27	0.834	0.96
DCT-CV	0.855	7.953	0.875	7.272	0.829	0.96
IM	0.856	7.956	0.883	7.284	0.848	0.961
CSR	0.86	7.57	0.888	7.294	0.863	0.961
PA-DCPCNN	0.869	8.215	0.896	7.346	0.867	**0.964**
LG-MW	0.853	8.057	0.888	7.296	0.864	0.96
QT	0.855	8.068	0.892	7.28	0.864	0.96
MST-SR	0.867	8.122	0.896	7.311	0.869	**0.964**
DSIFT	0.855	8.071	0.891	7.291	0.863	0.96
CNN	0.858	7.957	0.89	7.28	0.863	0.961
NSC -PCNN	0.86	7.822	0.887	7.298	0.849	**0.964**
Proposed Method	**0.954**	**9.424**	**0.989**	**7.389**	**0.923**	0.961

Table 3. The outcomes for the lab image and comparisons with existing techniques in the literature ([21]).

Evaluation Metric	*AG_F*	*Q_E*	*$Q^{AB/F}$*	*CC*	*SSIM*	*E(F)*
BF	6.456	0.863	0.895	0.977	0.907	6.989
BRW-TS	6.482	0.865	0.896	0.977	0.907	7.075
LG-MW	6.514	0.862	0.892	0.977	0.905	7.037
SSDI	6.539	0.865	0.897	0.977	0.907	7.096
IM	6.484	0.861	0.893	0.977	0.904	7.056
CSR	6.126	0.864	0.896	0.978	0.909	7.043
QT	6.504	0.865	0.896	0.977	0.906	7.039
GF	6.425	0.867	0.898	0.978	0.91	7.06
DSIFT	6.534	0.864	0.896	0.977	0.906	7.074
CNN	6.457	0.865	0.896	0.977	0.907	7.022
PA-DCPCNN	6.647	0.868	0.9	**0.979**	0.912	7.118
MST-SR	6.594	0.868	0.9	0.979	0.911	7.11
NSC -PCNN	6.336	0.85	0.893	0.979	0.909	6.992
DCT-CV	6.466	0.863	0.893	0.977	0.906	6.982
GFDF	6.479	0.865	0.896	0.977	0.907	7.032
Proposed Method	**7.599**	**0.917**	**0.993**	0.977	**0.965**	**7.153**

Table 4: The outcomes for the flowerpot image and comparisons with existing techniques in the literature ([22]).

Evaluation Metric	*Q_W*	*Q_E*
CNN-DWT	**0.9281**	0.8748
CNN	0.925	0.8722
DCTCV	0.9233	0.8692
IMF	0.9217	0.8679
MWGF	0.9215	0.8699
DS	0.9233	0.87
DWT	0.907	0.8603
GFF	0.925	0.8723
LP	0.9147	0.8687
Curvelet	0.925	0.8699
Proposed Method	0.9224	**0.9134**

4. CONCLUSION

DTCWT-based wavelet transform algorithm is proposed for the image fusion process which preserves directionality and shift invariant properties of the fused image. The implementation of filter architecture was carried out using the qshiftN algorithm. The combination of qshiftN and DTCWT architecture in the MR domain shows a greater advantage in the image fusion process than traditional wavelet algorithms such as DWT. The MPCA algorithm provides an additional advantage to the proposed algorithm to extract the features of the fused image. The performance of the proposed methodology is evaluated using various standard multifocus images and compared with the state-of-the-art algorithms reported recently. The visual and objective analyses were performed using different statistical metrics. The proposed algorithm based on DTCWT with qshiftN using MPCA shows its identity in terms of statistical metrics and showed its superior performance than compared with other standard fusion algorithms.

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